Monitoring Information Beyond Hazard Grades and Shift Type

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Question and matched comparison

This study uses the audited UCI Seismic Bumps mirror containing 2,578 records. A hazardous-shift label indicates a seismic bump with energy above 10,000 J in the subsequent eight-hour shift. It identifies a hazardous shift, not an observed accident or an avoided accident. Recorded order is a temporal proxy because timestamps and working-face identifiers are unavailable.

The primary seismic assessment orders a (no hazard), b (low hazard), c (high hazard) and d (danger state). The mirror contains only a, b for this field; c and d are defined levels but are not observed here. Average precision (AP) is the sum of precision values weighted by recall increments, evaluating tied scores as a group. It selects historical model configurations and is distinct from the detection counts evaluated at fixed inspection slots.

Does monitoring information improve hazardous-shift detection beyond three existing hazard assessments and shift type, at identical inspection allocations? This separately specified exploratory supplement compares a four-input base design with the established 17-column design. Both use XGBoost and the same four historical fitting/test splits. Each design independently selects depth 2/3 and 80/160 rounds on the same inner historical split by AP, with negative Brier if that inner set has no positives. Policy-reference and test labels do not enter fitting or tuning; no holdout is added.

Within each test block, K=floor(B*N) slots go to higher seismic grades first; model scores rank records within each grade. Both models use uniform allocation within an exactly tied boundary score group: h slots among m records containing p hazards contribute the boundary expectation below. Total expected detections add TP from all fully selected grade-score groups. Without a partially selected group, the boundary contribution is zero. Pooled total expectations sum block totals; scores are never ranked across fitted blocks. Fractional counts integrate allocation randomness.

$$E(TP_{boundary}) = \frac{hp}{m}$$

$$E(TP_{total}) = TP_{fully\ selected} + E(TP_{boundary})$$

Expected detections on the shared test records

The four blocks contain 516, 515, 516 and 516 records and 36, 10, 24 and 18 hazardous labels, respectively: 2,063 records and 88 labels in total. At the 20% budget, the pooled extended design detects 32.00 versus base 30.71 expected hazards with 412 alerts each; the difference is 1.29 [-5.48, 7.05]. All pooled intervals across the three specified block lengths include zero.

Table 1. Matched information designs at identical total and seismic-grade allocations. TP and FN are analytic expectations; K is the exact alert count. Positive differences favor the extended design. Brackets are conditional 95% intervals at block length 32.

Block	Budget	K	Base E(TP) / E(FN)	Extended E(TP) / E(FN)	Delta E(TP) [95% CI]
1	1%	5	0.94 / 35.06	0.00 / 36.00	-0.94 [-1.53, 0.17]
1	5%	25	4.69 / 31.31	5.00 / 31.00	0.31 [-4.46, 2.42]
1	10%	51	9.56 / 26.44	11.00 / 25.00	1.44 [-4.65, 3.25]
1	20%	103	16.47 / 19.53	17.00 / 19.00	0.53 [-1.41, 4.10]
2	1%	5	0.25 / 9.75	0.00 / 10.00	-0.25 [-0.40, 0.00]

Block	Budget	K	Base E(TP) / E(FN)	Extended E(TP) / E(FN)	Delta E(TP) [95% CI]
2	5%	25	1.23 / 8.77	0.00 / 10.00	-1.23 [-1.88, 0.48]
2	10%	51	2.51 / 7.49	0.00 / 10.00	-2.51 [-3.25, 1.13]
2	20%	103	5.17 / 4.83	5.00 / 5.00	-0.17 [-3.95, 1.55]
3	1%	5	0.30 / 23.70	0.00 / 24.00	-0.30 [-0.50, 0.82]
3	5%	25	1.48 / 22.52	3.00 / 21.00	1.52 [-1.23, 3.93]
3	10%	51	3.03 / 20.97	4.00 / 20.00	0.97 [-2.07, 5.09]
3	20%	103	6.11 / 17.89	6.00 / 18.00	-0.11 [-4.00, 4.14]
4	1%	5	0.14 / 17.86	1.00 / 17.00	0.86 [-0.32, 1.79]
4	5%	25	0.72 / 17.28	1.00 / 17.00	0.28 [-0.91, 2.55]
4	10%	51	1.47 / 16.53	2.00 / 16.00	0.53 [-1.63, 3.81]
4	20%	103	2.96 / 15.04	4.00 / 14.00	1.04 [-1.49, 3.66]
Pooled	1%	20	1.62 / 86.38	1.00 / 87.00	-0.62 [-2.20, 0.67]
Pooled	5%	100	8.12 / 79.88	9.00 / 79.00	0.88 [-4.95, 4.68]
Pooled	10%	204	16.56 / 71.44	17.00 / 71.00	0.44 [-5.40, 7.26]
Pooled	20%	412	30.71 / 57.29	32.00 / 56.00	1.29 [-5.48, 7.05]

Block-length sensitivity

The saved scores, tuning choices and models are frozen. Each of 2,000 paired moving-block draws reconstructs a queue separately within each block, recomputes grade allocations and reranks both score sets, including analytic tie expectations. The same sampled records feed both designs and every budget. Zero-positive draws are retained. Length 32 is primary; lengths 16 and 64 are displayed without selecting one from the outcomes. These are record-order blocks, not verified physical time durations. Intervals exclude fitting, tuning and selection uncertainty.

Discussion

The matched baseline changes the interpretation of the earlier complete-score-versus-grade-random pilot: the present increment is specifically relative to a fitted model already using all three assessments and shift type. Positive point differences occur in 3/4, 0/4, 2/4, 4/4 budgets in blocks 1-4, respectively; 16/16 main block-specific intervals include zero. The pooled intervals also include zero at every budget and block length, so these observations do not resolve a consistent monitoring-information increment under the specified workflow. This does not establish that monitoring information is generally uninformative: the comparison concerns these inputs, separately AP-selected configurations and previously examined records. At fixed hazardous count P and allocated slots K, expected loss uses the total expected detections as shown below; an additional cost grid would restate detection differences. Recorded order remains a temporal proxy, and the batch allocation is not an observed inspection process or accident-prevention effect. Independent timestamped working-face data with inspection outcomes would test transfer beyond this cohort.

$$E(L) = rP + K - (r+1)E(TP_{total})$$

Protocol and computational evidence

The protocol, fitting and allocation code, numerical verifier and saved evidence are linked below. The verifier recomputes the calculations through a separate path and supports full refitting replay. Retained

results include row-level scores and inclusion probabilities, all candidate tuning predictions, eight final models, grade allocations, every bootstrap multiplicity and replicate, and version-specific verification. The original stage-specific papers and ranking-pilot experimental results are retained. The integrated report includes the main matched-information finding; this supplement retains the complete design, table and block-length sensitivity.

Scientific code and saved-evidence links identify commit de83319. The integrated-manuscript link identifies the fixed presentation edition manuscript-2026-10-26. This revised report is generated by the current reporting code; its presentation is not claimed to be part of the earlier scientific source edition.

UCI dataset and target definition | Audited source and row mapping

Matched-information protocol | Model fitting and within-grade allocation

Numerical verification and refit replay | Saved results and verification guide

Integrated manuscript in this edition | Author contributions and AI assistance

Increment and interval figure

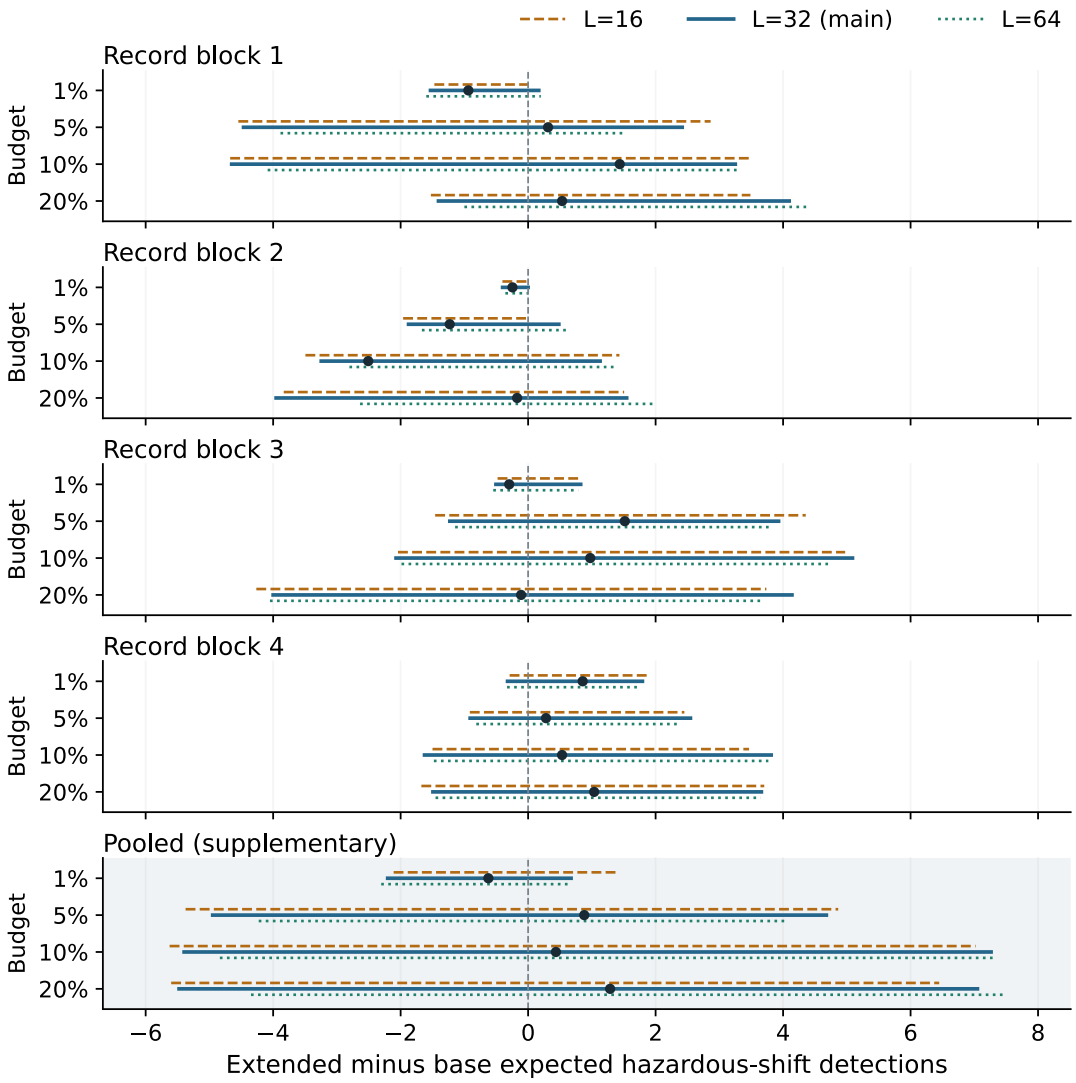


Figure 1. Extended minus base expected detections at identical allocations. Dots are the unchanged original-queue point differences. Lines show paired 95% intervals for lengths 16 (orange dashed), 32 (blue solid, main) and 64 (green dotted). Positive favors added monitoring. The shaded pooled panel is supplementary; block-specific evidence is primary.